

MULTIVARIATE LIST OF VARIABLES

Dependent Variables

Length of confinement: The length of confinement imposed (including any alternative incarceration), with a cap of 470 months (for example, a sentence imposed of life was coded as 470). The logarithm was used, with all sentences of zero months given the value of 0.01 months as the logarithm of zero is not mathematically possible. This variable was used for the ordinal least squares analyses. The variable used was SENSPLT0.

Non-Government Below Range Sentence: This variable was used for the “Likelihood of Obtaining a Non-Government Sponsored Below Range Sentence.” The variable BOOKERCD was used. This variable was coded “1” if the value for this variable was “downward departure,” “downward departure with *Booker*,” “below range with *Booker*,” and “remaining below range.”

Independent Variables

Guideline minimum: The minimum sentence, in months, the offender was subject to, taking into account all guideline, statutory and mandatory minimums. The logarithm of this variable was used, with all minimums of zero months given the value of 0.01 months, as the logarithm of zero is not mathematically possible. Minimums of life imprisonment were coded as 470 months. The variable used was GLMIN.

Type of offense committed (the variable used was GDLINEHI):

Violent offense: Offenders whose guideline sentence was controlled by the following guidelines were considered violent offenders: Chapter Two Part K offenders (“Offenses Involving Public Safety”), USSG §§2A1.1-2A1.5, 2A2.1-2A2.4, 2A4.1-2A4.2, 2A5.1-2A5.3, 2A6.1, 2A6.2, 2E1.3, 2E1.4, 2E2.1, 2B3.1, 2B3.2, and 2B3.3.

Sexual abuse offense: Offenders whose guideline sentence was controlled by the following guidelines: USSG §§2A3.1-2A3.4, and 2G1.1-2G1.3.

Pornography offense: Offenders whose guideline sentence was controlled by the following guidelines: USSG §§2G2.1-2G2. Drug trafficking offense: Offenders whose guideline sentence was controlled by the following guidelines: USSG §§2D1.1, 2D1.2, 2D1.5, 2D1.6, 2D1.7, 2D1.8, 2D1.9, 2D1.10, 2D1.11, 2D1.12, and 2D1.13.

White collar offenses: Offenders whose guideline sentence was controlled by the following guidelines: USSG §§2B1.1, 2B1.6, 2B4.1, 2B5.1, 2B5.3, 2F1.1, 2F1.2, 2R1.1, Chapter Two Part S offenses (“Money Laundering and Monetary Transaction Reporting”), and Chapter Two Part T offenses (“Offenses Involving Taxation”).

Immigration offenses: Offenders whose guideline sentence was controlled by offenses in Chapter Two Part L (“Offenses Involving Immigration, Naturalization and Passports”).

Other type offenses: Offenders whose guideline sentence was controlled by offenses not in the violent, sexual abuse, pornography, drug trafficking, white collar offenses, and immigration offenses.

Sentence placement relative to guideline range: Prior to the *Booker* decision, this was more commonly referred to as “departure status.” Pre-*Booker* the variable used was DEPART. Upward departures were coded as “above range sentence”, downward departures were coded as “below range sentence” and substantial assistance departures retained their identity. Post-*Booker*, the variable BOOKERCD was used. “Above range sentence” was coded if the values were “upward departure - guideline reason,” “upward departure - guideline and 18 U.S.C. §3553 (3553) reason,” “above range with *Booker* and 3553 reason,” and “other above range.” “Below range sentence” was coded if the values were “downward departure - guideline reason,” “downward departure - guideline and 3553 reason,” “below range with *Booker* and 3553 reason,” “other below range,” “other below range,” “early disposition/§5K3.1” and “government sponsored - below range.” Substantial assistance departures retained their identity.

Trial: Whether the offender was tried either by jury or by the court. The variable used was NEWCNVTN.

Mandatory minimum application: If the statutory minimum for the offense was greater than zero, a mandatory minimum applies in the case. If the sentencing court was relieved from application of the mandatory either via a substantial assistance departure or application of safety valve, the mandatory minimum does not apply and this variable was coded as “0.” The variables used were STATMIN, BOOKERCD, DEPART and SAFE.

Pre-sentence custody status: The offender’s pre-sentence detention status. The variable used was PRESENT.

Race and gender of offender: The variables NEWRACE (race or ethnicity of the offender) and MONSEX (gender of the offender) were merged to form one value for each offender.

Age of offender: This was coded as a dichotomous variable, separating those who were 25 years of age and younger from those older than 25 years of age. The variable used was AGE.

Educational attainment: This was coded as a dichotomous variable, separating those offenders who attended college for any period of time and those who never attended college. The variable used was EDUCATN.

Citizenship: The variable NEWCIT was used.

Drug Type: This variable was used in the “Multivariate Analysis - Drug Trafficking” analysis. Only cases that involved cocaine, crack, heroin, marijuana and methamphetamine was used. The variable COMBDRG2 was used.

Refined Model
Post-PROTECT

The REG Procedure
Model: MODEL1
Dependent Variable: logsplit

Number of Observations Read 63750
Number of Observations Used 63750

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	23	262379	11408	7006.62	<.0001
Error	63726	103755	1.62814		
Corrected Total	63749	366133			

Root MSE 1.27599 R-Square 0.7166
Dependent Mean 2.85964 Adj R-Sq 0.7165
Coeff Var 44.62052

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Standardized Estimate
Intercept	1	0.61972	0.02795	22.17	<.0001	0
logmin	1	0.70087	0.00254	275.64	<.0001	0.73059
drugtraff	1	0.13157	0.01707	7.71	<.0001	0.02685
sexual2	1	0.14899	0.06296	2.37	0.0180	0.00516
porn	1	0.12196	0.05021	2.43	0.0151	0.00539
immigration	1	0.19885	0.02163	9.19	<.0001	0.03356
othtype	1	-0.42711	0.02997	-14.25	<.0001	-0.03379
whitecoll	1	-0.35248	0.01971	-17.88	<.0001	-0.05714
upward	1	1.12157	0.04604	24.36	<.0001	0.05174
downdep	1	-0.93116	0.01614	-57.70	<.0001	-0.12854
subasst	1	-1.27297	0.01460	-87.19	<.0001	-0.19978
mandmin2	1	0.34250	0.01665	20.57	<.0001	0.05174
newcnvtn	1	0.21649	0.02471	8.76	<.0001	0.01899
custody	1	0.68759	0.01331	51.66	<.0001	0.13279
whitefemale	1	-0.26209	0.02455	-10.67	<.0001	-0.02446
blackmale	1	0.05346	0.01536	3.48	0.0005	0.00907
blackfemale	1	-0.42348	0.02813	-15.06	<.0001	-0.03398
hispmale	1	-0.04531	0.01689	-2.68	0.0073	-0.00909
hispfemale	1	-0.26309	0.02893	-9.09	<.0001	-0.02083
othermale	1	-0.08359	0.03026	-2.76	0.0057	-0.00621
otherfemale	1	-0.56858	0.05780	-9.84	<.0001	-0.02105
agedummy	1	0.01955	0.01242	1.57	0.1154	0.00340
educ	1	-0.03990	0.01338	-2.98	0.0029	-0.00680
citizen	1	-0.02114	0.01626	-1.30	0.1935	-0.00416

Refined Model
Post-Booker

The REG Procedure
Model: MODEL1
Dependent Variable: logsplit

Number of Observations Read 178418
Number of Observations Used 178418

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	23	644083	28004	16031.0	<.0001
Error	178394	311627	1.74685		
Corrected Total	178417	955709			

Root MSE 1.32168 R-Square 0.6739
Dependent Mean 2.95157 Adj R-Sq 0.6739
Coeff Var 44.77895

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Standardized Estimate
Intercept	1	0.55903	0.01786	31.30	<.0001	0
logmin	1	0.70384	0.00171	412.35	<.0001	0.70205
drugtraff	1	0.14012	0.01072	13.07	<.0001	0.02949
sexual2	1	0.05343	0.03813	1.40	0.1611	0.00196
porn	1	0.32204	0.02565	12.56	<.0001	0.01861
immigration	1	0.11072	0.01320	8.39	<.0001	0.02007
othtype	1	-0.47651	0.01875	-25.41	<.0001	-0.03900
whitecoll	1	-0.37897	0.01255	-30.21	<.0001	-0.06158
upward	1	1.32065	0.02210	59.76	<.0001	0.08187
downdep	1	-0.90906	0.00795	-114.34	<.0001	-0.16695
subasst	1	-1.16007	0.00956	-121.39	<.0001	-0.18196
mandmin2	1	0.32716	0.00989	33.06	<.0001	0.05362
newcnvtn	1	0.30897	0.01511	20.45	<.0001	0.02841
custody	1	0.72786	0.00852	85.41	<.0001	0.14086
whitefemale	1	-0.21120	0.01563	-13.51	<.0001	-0.01993
blackmale	1	0.14130	0.00968	14.59	<.0001	0.02508
blackfemale	1	-0.28593	0.01858	-15.39	<.0001	-0.02226
hispmale	1	-0.00996	0.01071	-0.93	0.3523	-0.00209
hispfemale	1	-0.26311	0.01757	-14.98	<.0001	-0.02221
othermale	1	0.02567	0.01832	1.40	0.1611	0.00202
otherfemale	1	-0.25818	0.03473	-7.43	<.0001	-0.01023
agedummy	1	0.03029	0.00797	3.80	0.0001	0.00526
educ	1	-0.05544	0.00844	-6.57	<.0001	-0.00971
citizen	1	-0.08864	0.01003	-8.84	<.0001	-0.01823

MULTIVARIATE ANALYSIS RESULTS

Multivariate Analysis - All Cases
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The REG Procedure
Model: MODEL1
Dependent Variable: logsplit

Number of Observations Read 261407
Number of Observations Used 261407

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	23	859974	37390	17260.1	<.0001
Error	261383	566230	2.16628		
Corrected Total	261406	1426204			

Root MSE	1.47183	R-Square	0.6030
Dependent Mean	2.78314	Adj R-Sq	0.6029
Coeff Var	52.88385		

Parameter Estimates

Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Standardized Estimate
Intercept	1	0.64136	0.01718	37.33	<.0001	0
logmin	1	0.63193	0.00149	422.89	<.0001	0.65832
drugtraff	1	0.06341	0.01035	6.13	<.0001	0.01291
sexual2	1	0.07269	0.03672	1.98	0.0478	0.00252
porn	1	0.44969	0.02123	21.18	<.0001	0.03027
immigration	1	0.05162	0.01243	4.15	<.0001	0.01011
othtype	1	-0.74296	0.01800	-41.27	<.0001	-0.05816
whitecoll	1	-0.42280	0.01213	-34.86	<.0001	-0.06529
upward	1	1.14187	0.01903	60.01	<.0001	0.07504
downdep	1	-0.97350	0.00692	-140.75	<.0001	-0.19323
subasst	1	-1.17370	0.00954	-123.09	<.0001	-0.17108
mandmin2	1	0.50636	0.00936	54.10	<.0001	0.08047
NEWCNVTN	1	0.42336	0.01580	26.79	<.0001	0.03383
custody	1	0.92891	0.00827	112.35	<.0001	0.17009
whitefemale	1	-0.37270	0.01470	-25.36	<.0001	-0.03427
blackmale	1	0.17781	0.00942	18.88	<.0001	0.02987
blackfemale	1	-0.40153	0.01863	-21.55	<.0001	-0.02834
hispmale	1	0.00580	0.00958	0.61	0.5448	0.00123
hispfemale	1	-0.20146	0.01603	-12.57	<.0001	-0.01702
othermale	1	0.00043741	0.01792	0.02	0.9805	0.00003193
otherfemale	1	-0.42400	0.03480	-12.18	<.0001	-0.01525
agedummy	1	0.02490	0.00755	3.30	0.0010	0.00415
educ	1	-0.04681	0.00806	-5.81	<.0001	-0.00794
citizen	1	-0.10983	0.00950	-11.56	<.0001	-0.02316

Appendix D: Regression Model Output

Regression model 2012-2016							The REG Procedure Model: MODEL1 Dependent Variable: logsplit		
			Number of Observations Read		322275				
			Number of Observations Used		322275				
Analysis of Variance									
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F				
Model	23	1068533	46458	18451.7	<.0001				
Error	322251	811366	2.51781						
Corrected Total	322274	1879899							
Root MSE									
Dependent Mean									
Coeff Var									
1.58676							R-Square		
2.68470							Adj R-Sq		
59.10388							0.5684		
0.5684							0.5684		
Parameter Estimates									
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Standardized Estimate			
Intercept	1	0.21375	0.01371	15.59	<.0001	0			
logmin	1	0.61526	0.00148	415.49	<.0001	0.61552			
drugtraff	1	0.18545	0.00959	19.33	<.0001	0.03604			
sexual2	1	0.23591	0.02985	7.90	<.0001	0.00949			
porn	1	0.62975	0.01917	32.85	<.0001	0.04464			
immigration	1	-0.03436	0.01217	-2.82	0.0048	-0.00647			
othtype	1	-0.57546	0.01733	-33.21	<.0001	-0.04382			
whitecoll	1	-0.20351	0.01154	-17.63	<.0001	-0.03100			
upward	1	0.64847	0.01957	33.13	<.0001	0.03897			
downdep	1	-1.06743	0.00662	-161.36	<.0001	-0.21512			
subasst	1	-1.11105	0.00992	-111.95	<.0001	-0.15554			
mandmin2	1	0.47247	0.00966	48.92	<.0001	0.06714			
NEWCONVTN	1	0.50739	0.01632	31.09	<.0001	0.03693			
custody	1	1.34406	0.00819	164.17	<.0001	0.24219			
whitefemale	1	-0.34097	0.01444	-23.61	<.0001	-0.03029			
blackmale	1	0.17520	0.00949	18.46	<.0001	0.02790			
blackfemale	1	-0.35241	0.01792	-19.66	<.0001	-0.02452			
hispmale	1	0.05176	0.00976	5.30	<.0001	0.01067			
hispfemale	1	-0.18347	0.01478	-12.41	<.0001	-0.01629			
othermale	1	-0.03887	0.01704	-2.28	0.0225	-0.00284			
otherfemale	1	-0.43742	0.03152	-13.88	<.0001	-0.01640			
agedummy	1	0.02821	0.00783	3.60	0.0003	0.00428			
educ	1	-0.00270	0.00765	-0.35	0.7241	-0.00045774			
NEWCIT	1	0.04916	0.00938	5.24	<.0001	0.00996			